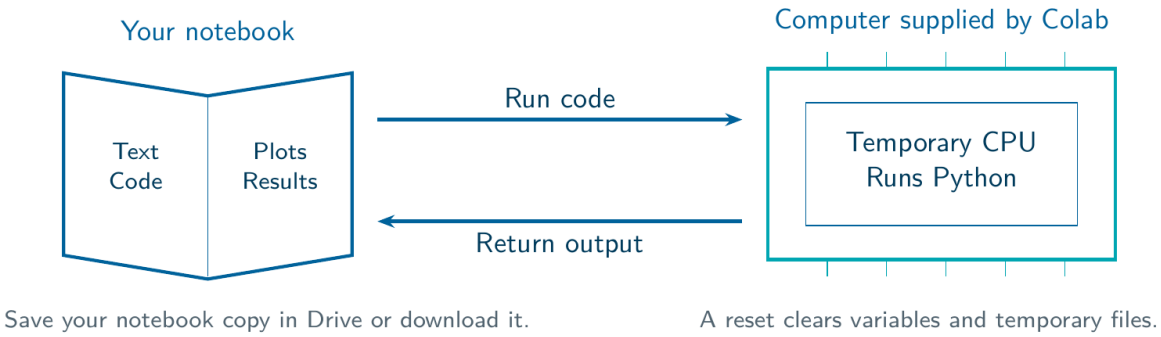
# Start with one notebook

Classroom Labs / participant guide




**Presenters:** Alaa Selim, MIT/Dartmouth; Mohamed Chehada, UT Austin.

Open **Classroom_Labs.ipynb**. It covers voltage, restoration, cyber and the saved Llama replays. The website's other notebooks are companion resources.

**1. Keep your copy.** Sign in and save a copy in Drive, or download your notebook when you finish.

**2. Connect a CPU.** Run the first setup cell. Wait for **Ready**, then run **L0** and **L1**. Keep both switches False.

**3. Follow the slides.** Run each named cell when prompted. You will change one number in each case. No coding experience or live LLM service is required.

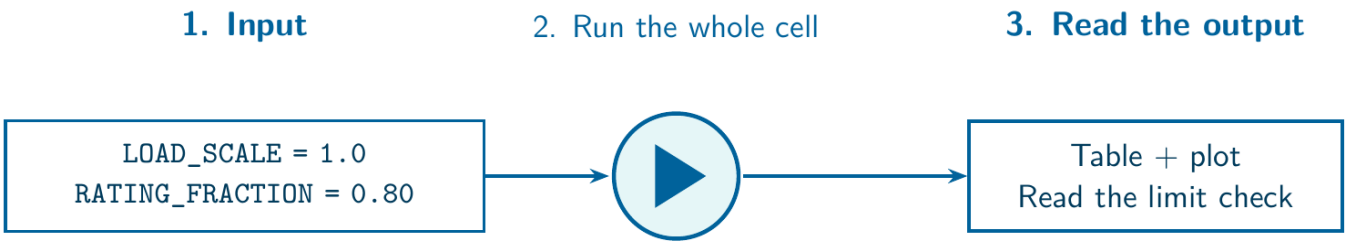
The runtime is a temporary computer. Saving the notebook does not keep its temporary files or variables. After a reset, rerun setup and earlier cells.

**Before the first case, show your partner where the code, play button and output appear.**

Open Classroom Labs in Google Colab
Scan the code or use the website's first Colab button.

# Learn with one small experiment

Predict / run / read / change / explain




The play button executes Python. The saved text below a cell is its output.

**Read the input.** Find the L heading in the notebook. The slide shows selected lines; run the full cell.

**Predict first.** Tell your partner what you expect before revealing the result.

**Run and wait.** Use the play button beside that cell. Read the new table or plot underneath it.

**Change one input.** Rerun the cell, compare outputs and explain which check supports your next step.

**Use the labels.** Fresh means calculated now. Recorded replay means an actual old model reply with its original context. A saved controller can produce a new proposal from old weights.

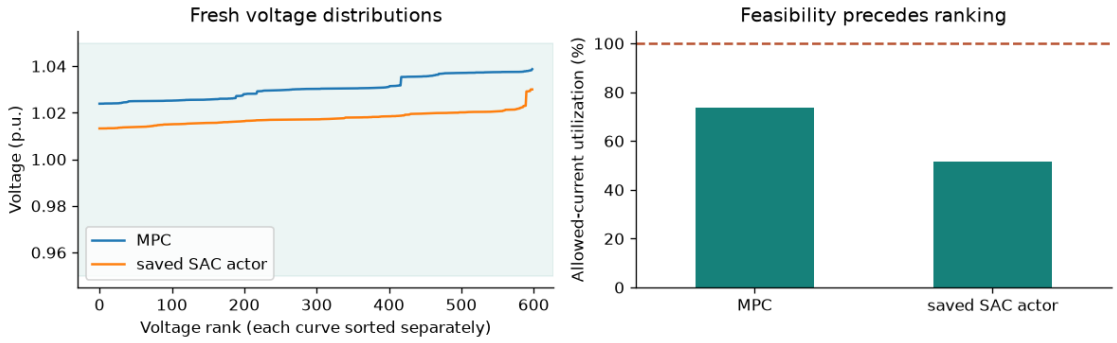
**Common messages:** HOLD means a required condition is missing or unmet. REVIEW means a person still needs to assess it. L11 SKIPPED is expected when the live-model switch is False.

**If a variable is missing:** rerun the required earlier cells. Editing a number does not update an old output until you rerun.

**Work in pairs: one person runs, one checks. Swap roles for the next case.**

# Case 1: voltage support

L2–L5 / exercise on slide 19




**Goal:** compare two control proposals, then check their voltage and current limits.

**Run L2 at the defaults:** LOAD_SCALE = 1.0 and RATING_FRACTION = 0.80. Read feasible, current_step_J and max_allowed_current_pct.

**Read the plots:** the left plot shows sorted modeled voltages. The right plot compares current use with the 100% allowed-current line.

**Four-minute task:** change only RATING_FRACTION to **0.75**. Predict, rerun all of L2 and compare both methods. Check limits before comparing the score J.

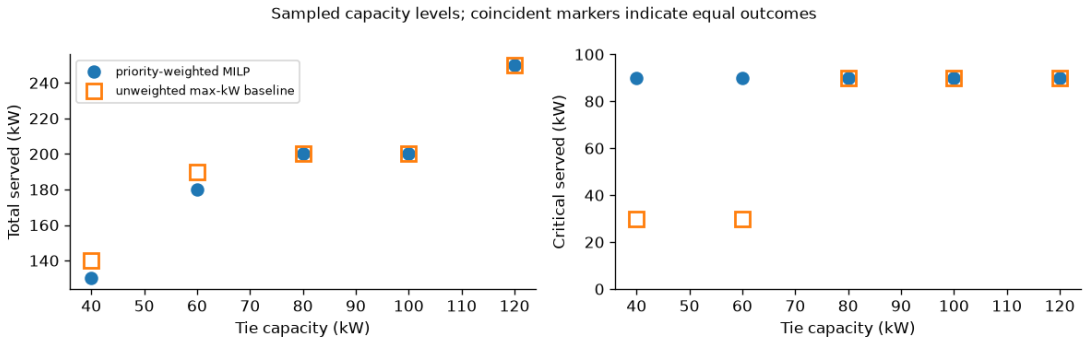
**Reset:** restore 0.80 and rerun L2. Then follow L3–L5. Those cells use the separate original test_01 archive; your edited L2 does not rewrite it.

**Model coverage:** the default check covers the MV model. It does not prove that every LV device or dynamic limit passes.

**Record: the changed allowance, each current percentage, pass/fail, and what the model does not cover.**

# Case 2: restoring service

L6–L7 / exercise on slide 39




**Goal:** see why restoring the most kW can differ from serving critical loads.

**Run L6:** compare served_kw and critical_kw for the priority-weighted plan and the maximum-kW plan. Both use the same six-node graph.

**Read the plots together:** total service is on the left; critical service is on the right. Explain the tradeoff at 40 kW.

**Four-minute task:** append **180.** to CAPACITIES_KW. Keep **40.** first. Predict, rerun L6 and compare the 40 kW and 180 kW outcomes.

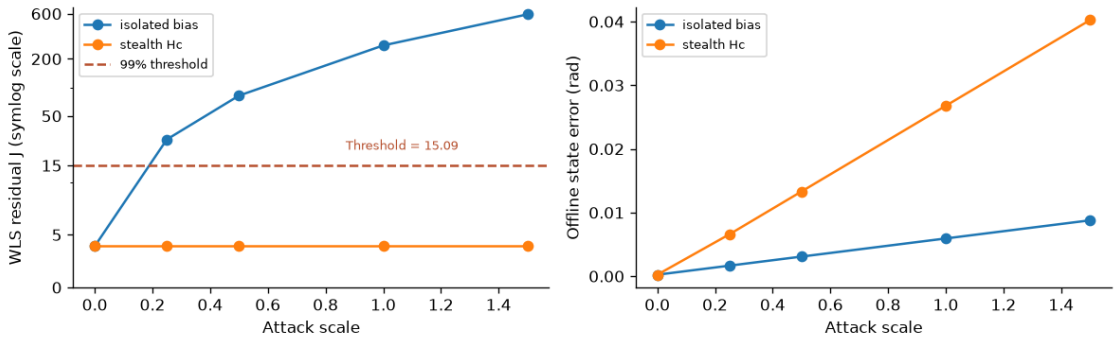
**Reset before L7:** restore the original capacity list and rerun L6. L7 uses the first record, then tests a changed request with the old evidence.

**Meaning of HOLD:** old evidence no longer matches. That does not prove a newly calculated plan would fail. The model also omits AC voltage and physical switching order.

**Record: total and critical kW at both capacities. Explain which service goal each plan follows.**

# Case 3: checking suspicious readings

L8–L9 / exercise on slide 55




**Goal:** distinguish a reading that fits the model from a reading that is true.

**Run L8:** compare isolated bias with coordinated Hc shifts. The residual alarm measures mismatch. State error uses known simulation truth for teaching.

**Four-minute task:** add **.10** to ATTACK_SCALES. Predict, rerun L8 and compare the alarm with the state error at that scale.

**Reset before L9:** restore the original scale list and rerun L8. L9 is a separate fixed recovery case; its correction does not follow your changed sweep.

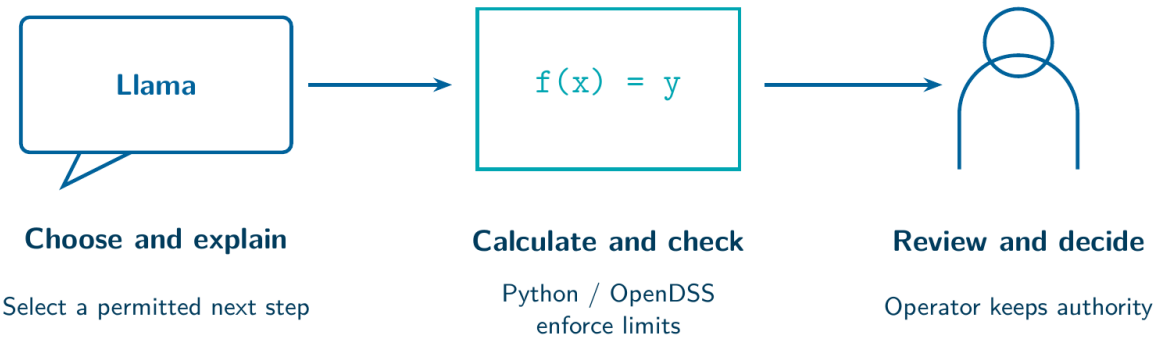
**Run L9:** no trusted replacement data gives HOLD. Assuming the additional readings are trusted allows a PROPOSE result. The calculation does not authenticate those readings.

**Discuss:** a small residual or clear model explanation alone cannot establish sensor trust.

**Record: residual J, alarm flag, state error and the condition needed before a recovery proposal.**

# Read Llama’s choices

L10 / L10B / L11 / L12




**The opening and case slides preview saved agent examples.** Follow the notebook in order; run the replays here after the three cases.

**Run L10, then L10B:** set SCENARIO = "voltage", then "restoration" and "cyber", rerunning each time. These use separate recorded studies.

**Follow the agentic loop:** goal, Llama tool choice, returned evidence, next choice. The replay makes no new model call.

**Restoration challenge:** find the wrong Llama ranking. Explain why the numerical guard keeps the lower-cost passed candidate.

**L11:** keep the live call disabled. **L12:** inspect the full-scope HOLD, the wrong-ID rejection and the simulated readback. equipment_dispatch stays False.

**Exit ticket:** explain one result using input, tool, result, check and operator decision. Save the changed inputs and outputs.

**Choose one case. Which decision did Llama make, which fact came from a tool, and what must the operator still review?**

Monica Lam. *CS224V Agentic AI, Lecture 1: Introduction.* Stanford University, Fall 2026, pp.19–20, 40 and 52–53. Full lecture PDF.
Teaching ideas adapted to our own grid examples.
Google Colaboratory. *Frequently Asked Questions.* research.google.com/colaboratory/faq.html.
Full paper citations, figure credits and exact notebook source map: Teaching and Source Guide.